Masters IT Project — ACU Sydney

Memory Companion Agent

An AI-powered web app that helps Alzheimer's patients have warm, personalised conversations — using their own life stories, photos, and memories. Caregivers get daily summaries automatically.

Agentic AI · 3 Specialised Agents · LangGraph · RAG + Memory · 10 Weeks · Browser-based

THE PROBLEM

Over **55 million** people live with dementia worldwide. Caregivers spend hours doing *reminiscence therapy* — talking through old memories to improve mood. There is no affordable AI system that does this automatically.

55M	59%
people with dementia	caregivers burnt out

OUR SOLUTION

A caregiver uploads a patient's **photos + biography** once. The AI then holds daily personalised conversations, detects if the patient is upset, redirects gently, and emails a **daily summary** to the family.

3	~$15
AI agents working together	total API cost

HOW IT WORKS — 3 AGENTS BEHIND EVERY CONVERSATION

Reminiscence Agent
Retrieves personal facts from the patient's uploaded biography and past sessions. Generates warm, grounded responses using only *real memories*.

Mood Agent
Reads every message for emotional tone — happy, confused, or distressed. If distress is detected, it signals the supervisor to redirect the conversation gently.

Report Agent
At session end, auto-generates a plain-English daily summary for the caregiver — topics discussed, mood trend, and any concerns to follow up on.

KEY FEATURES

- **Personalised conversations** — Based on patient's own photos and life story
- **Real-time mood detection** — Detects distress and redirects automatically
- **Cross-session memory** — Remembers what was said in previous sessions
- **Caregiver daily report** — Auto-generated summary emailed to family
- **Voice output (TTS)** — Reads responses aloud via browser — no typing
- **Simple caregiver setup** — Upload photos + bio once, done

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Margaret, 78 — Alzheimer's patient
Morning session · Bondi Beach memory

AI: Good morning Margaret! I was looking at that beautiful photo of you at Bondi Beach in 1964. You looked so happy — was that a family holiday?

Reminiscence Agent used biography RAG

Yes! We used to pack sandwiches and spend the whole day there with the kids...

AI: That sounds wonderful. Your daughter Sarah mentioned you made the best Vegemite sandwiches. Did David help pack the bag?

Mood: happy — Reminiscence Agent continues

Where is David? I want David.

AI: David sounds like he was such a wonderful part of your life. You know what — I have some of that folk music you both loved. Shall we listen together?

Mood: distressed — Supervisor redirected to calm mode

Session: 28 min	1 distress event	Report sent
Engaged & happy	Redirected in 3 sec	To daughter Sarah

TECH STACK

LangGraph · Claude API · LlamaIndex · ChromaDB · Mem0 · FastAPI · React · Web Speech API · Vercel

TEAM — BHARAT · VIVEK · ANISH · ROHIT

B	V	A	R
Bharat	Vivek	Anish	Rohit
AI & Backend	Data Pipeline	Frontend	UI & Testing